NATIONAL AI HACKATHON 2026 · SCHUBKI × ART FIBER SYSTEMS

AFS In-line Spectroscopy Assistant

From customer enquiry to first experiment: four AI agents draft the expert's first technical assessment, with every claim sourced and checked.



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afsnationalaihackathon.vercel.app

THE PROBLEM

Customers know what to measure. Not how.

Every enquiry needs an AFS expert to choose the method, spectral range, fiber and first test, and many rest on a wrong premise. Three from the evaluation set:

H04 · CLIENT PREMISE

Reuse the existing silica fiber for MIR ATR

Refuted. Silica fibers carry 190–2400 nm; MIR ATR needs 2.5–25 µm.

S02 p.6

H06 · CLIENT PREMISE

Quote an in vitro study as proof for surgery

Refuted. A 14-sample in vitro proof of concept is not clinical evidence.

S03 p.1, p.14

H03 · CLIENT PREMISE

Fluorescence is very sensitive, so it fits

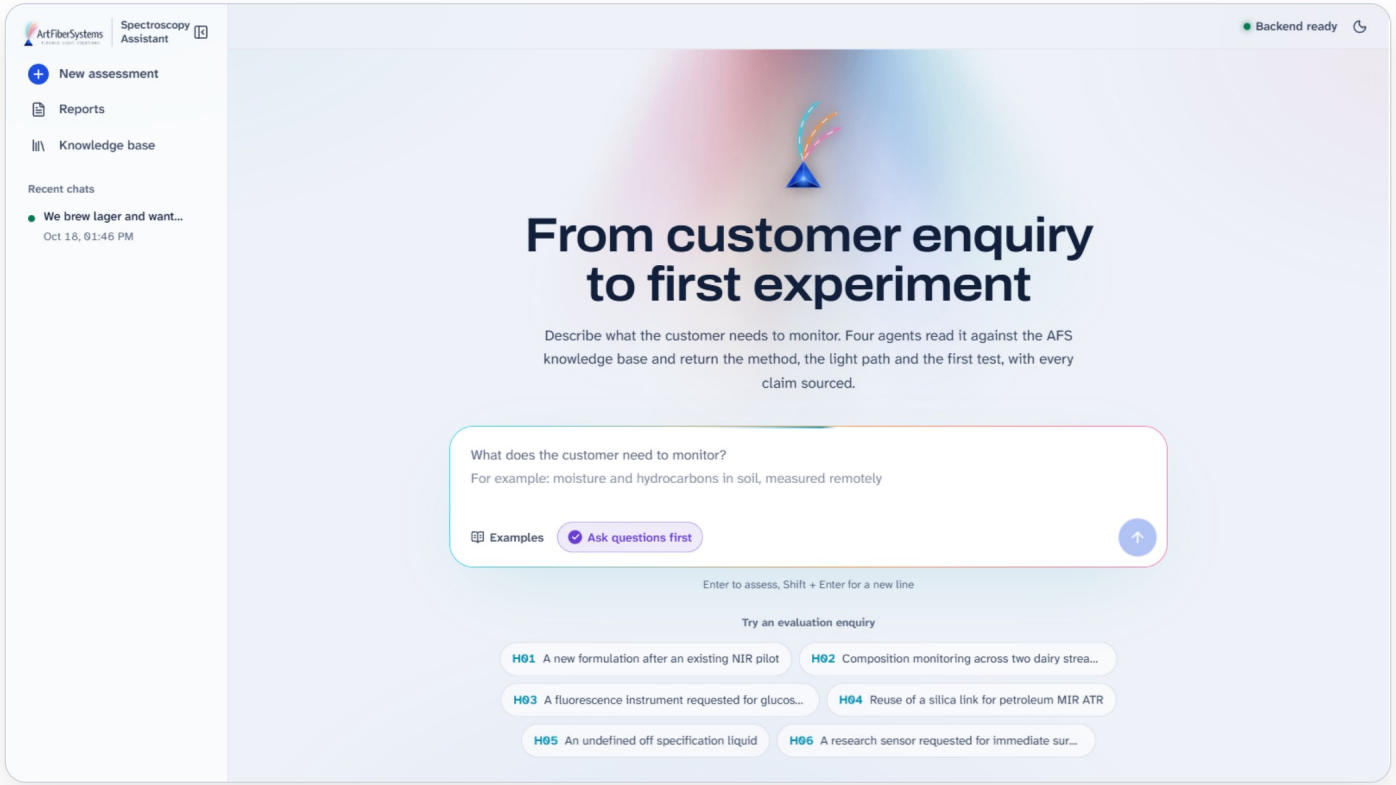
Refuted. Sensitivity is not selectivity; fluorescence needs a fluorescent target.

S01 Table 1, p.16

The hard part is not answering. It is catching the wrong premise, with the page that proves it.

WHAT WE BUILT

Enquiry in. First assessment out.



EVERY REPORT CONTAINS

01 Requirements: facts vs. unknowns

02 Prioritised clarifying questions

03 A verdict on every client premise

04 Methods to test, and to avoid

05 Spectral range, in correct units

06 Probe and fiber concept

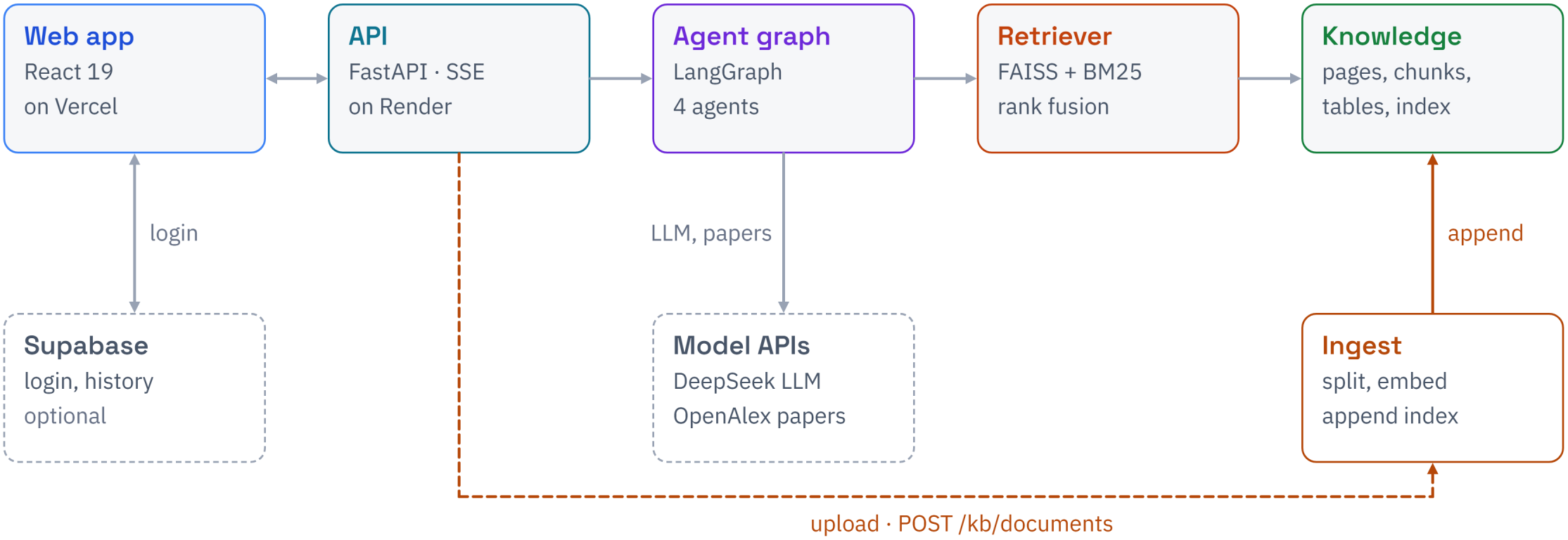
07 Risks and a decision status

08 A practical first experiment

Built to be reviewed: every claim is typed, and every citation opens its source page.

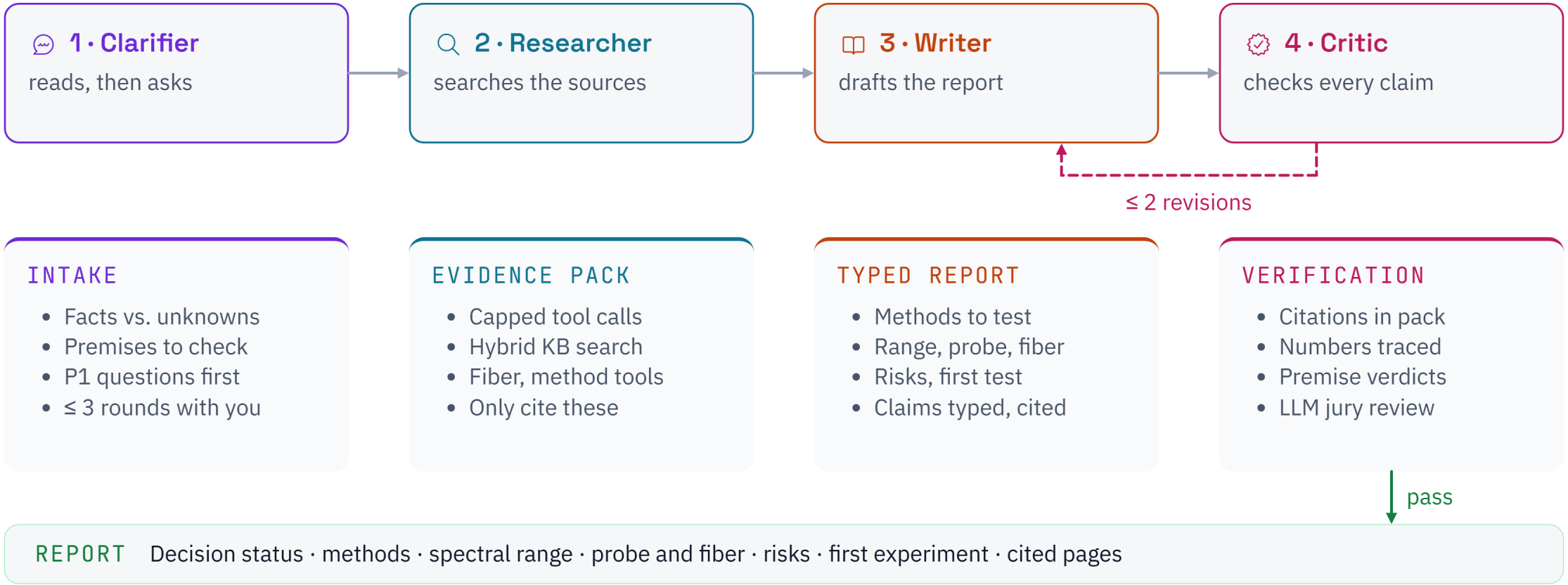
ARCHITECTURE · SYSTEM

System architecture, end to end



ARCHITECTURE · AGENT FLOW (LANGGRAPH)

Agent flow: four agents, one evidence trail



The AFS materials are the evidence

ID	Source	Evidence level	Pages
S01	**Multispectral Laboratory of Optical Analysis Methods: purpose, methods and applications** Overview of optical analysis methods and application examples. Main technical source. Methods p.2–6 (comparison table p.6), combining methods p.7–9, measurement kits and probes p.10–12, application cases p.13–18 (medical, pharmaceutical, bioprocess, dairy).	Vendor overview	19
S02	**Process-Control in-line with Spectral Fiber Systems (AFS slides)** AFS slides on process control: fiber types and transmission ranges (p.6–10), probes, application slides (water in petroleum p.13, API in polymer extrusion p.14), combined probes (p.15–16). Mostly images: text extraction returns little, so read the slides visually.	Vendor slides	20
S03	**Development and Testing of an LED-Based Near-Infrared Sensor for Human Kidney Tumor Diagnostics (Sensors, 2017)** Research paper: LED-based NIR sensor for kidney tumour diagnostics, tested on 14 in vitro samples from 7 patients. A proof of concept.	In vitro proof of concept	17
S04	**Fiber Probe for Simultaneous Mid-Infrared and Fluorescence Spectroscopic Analysis (Analytical Chemistry, 2021)** Research paper: fiber probe for simultaneous MIR (ATR) and fluorescence measurement, demonstrated on model biological tissues.	Lab research note	6
S05	**Art Fiber Systems flyer** Company and product overview. Business context, not evidence of technical performance.	Business context	2
S06	**MIR Photonics laboratory website link** Link to the laboratory website (https://mirphotonics.com/laboratory/). Business context.	Business context	1

READ PAGE BY PAGE

All 20 image-only S02 slides and 9 S01 figure pages read by a vision model; Table 1 from its clean transcription.

CITED TO THE PAGE

Search chunks never cross a page, so every hit is a citation like S02 p.6 that opens that exact page.

SCOPE TRAVELS WITH IT

Each page carries its source’s evidence level: S03 is an in vitro proof of concept, never clinical evidence.

COMPUTED FROM THE SOURCES

S01 Table 1 and the AFS fiber ranges are structured tools, so fiber–method fit is calculated, not guessed.

EXAMPLE · EVALUATION CASE H01 · SUMMARISED FROM outputs/H01.md

Recommendation and first feasibility plan

THE ENQUIRY

NIR pilot at a tablet-press feed frame, new API blend: reuse the calibration and add Raman to keep it valid?

BOTH PREMISES REFUTED

Reuse the calibration: a new blend is a new matrix, so the NIR model must be rebuilt and revalidated. S01 p.10, p.14

Raman keeps it valid: adding Raman builds a new fused model, not a repair of the old one. S01 p.15, p.7

RECOMMENDATION · PROVISIONAL

Keep the feed-frame NIR diffuse-reflection set-up, with a new calibration for the new blend. S01 p.14, Table 1

Compare Raman at 785 nm; fuse NIR and Raman only if NIR alone falls short. S01 p.15, p.7

FEASIBILITY PLAN

SAMPLES	REFERENCE
New blend across the agreed API range	Chromatography or mass spectrometry

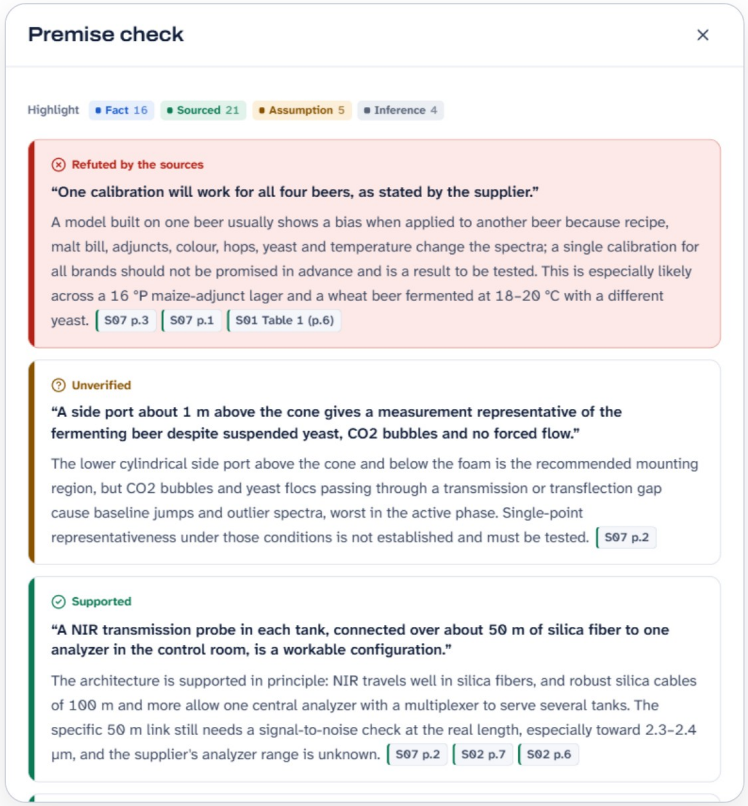
1. Scope the old calibration and the probe and fiber link
2. Lab screening of the new blend with NIR spectra
3. Test the old calibration against reference values
4. Add Raman or fused NIR + Raman only if NIR falls short
5. Feed-frame trial against timed reference samples

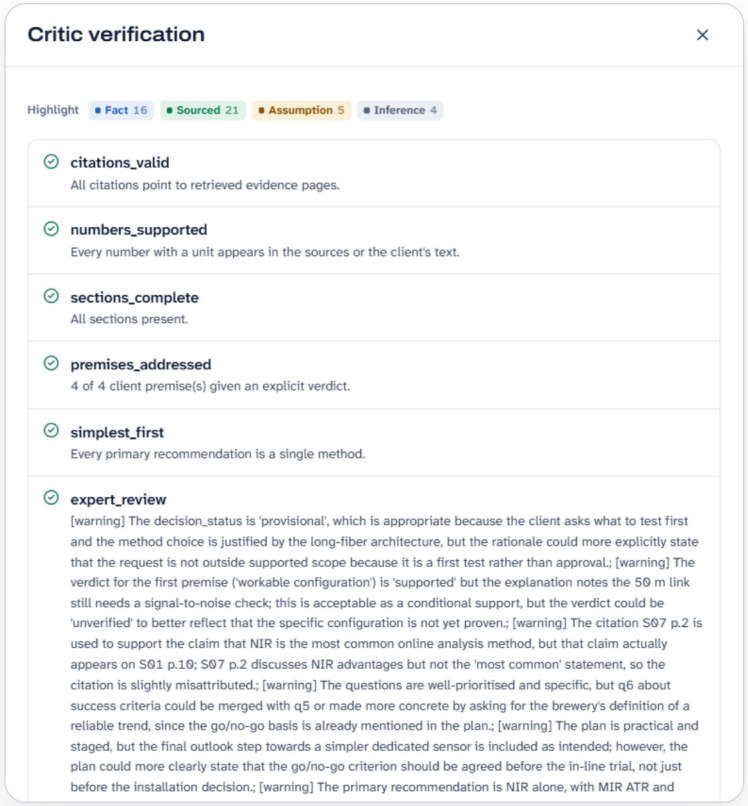
GO / NO-GO

Go if a model meets the client's error, bias and response-time limits on independent feed-frame samples.

TRUST · ENFORCED IN CODE

Every claim typed, sourced and checked





NO CITATION OUTSIDE THE PACK

Only pages retrieved in this run can be cited; each opens its exact page.

NO NUMBER WITHOUT A SOURCE

Every value with a unit is traced; nm, µm and cm^{-1} count as one.

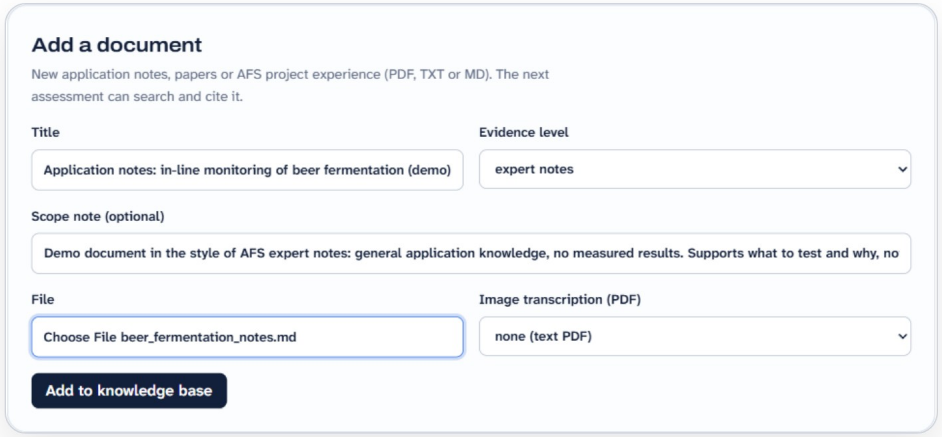
NO PREMISE WITHOUT A VERDICT

Supported, unverified or refuted, with the pages behind each.

4 CLAIM TYPES

Fact, sourced, inference, assumption. In the live run shown: 16 · 21 · 4 · 5.

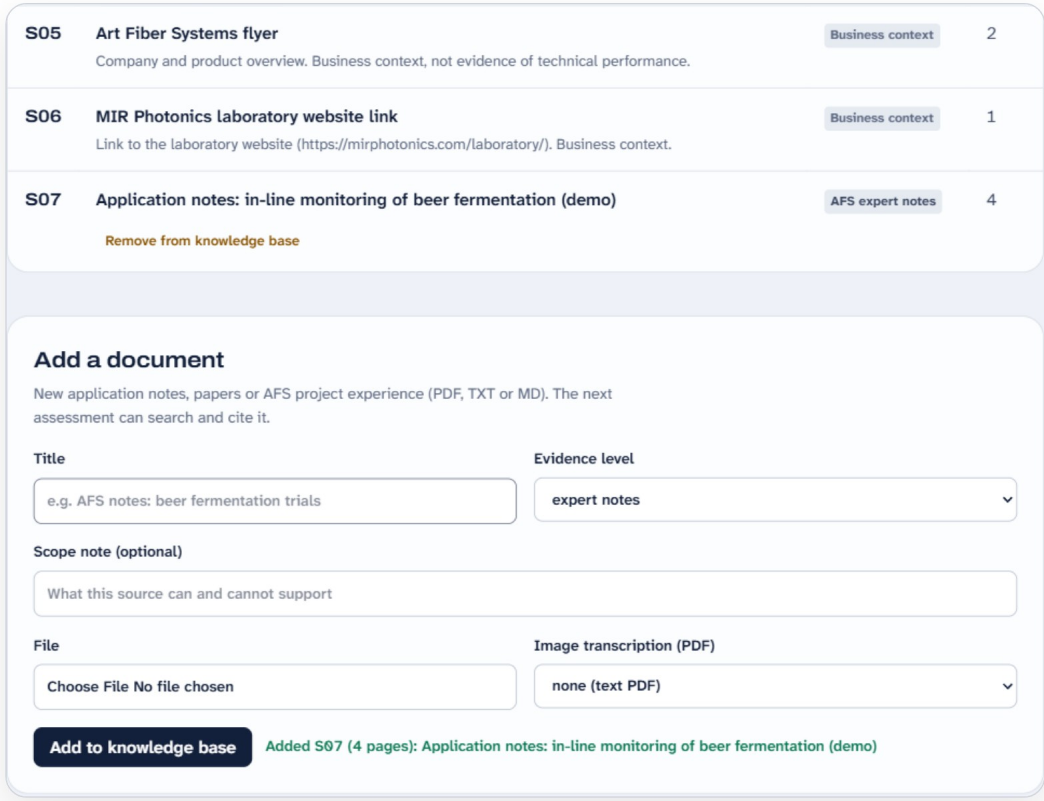
Add a document; the next run cites it



01 Upload a PDF, TXT or MD with its evidence level and scope note: from the UI, the API or the CLI.

02 It becomes S07: only its new chunks are embedded and appended. No rebuild, no retraining.

03 The next assessment can cite it: our live run cited S07 p.1–4 minutes later.



PROOF · 14 REPORTS · BATCH MODE · NO MANUAL EDITS

Every report passed every check

14/14
reports pass all critic checks (D01–D08, H01–H06)

770
citations to AFS pages, every one valid

0
unsupported numbers across 382 claims

47
client premises given an explicit verdict

THE SIX EVALUATION ENQUIRIES

H01	H02	H03	H04	H05	H06
Provisional	Provisional	Outside supported scope	Outside supported scope	Outside supported scope	Outside supported scope
new formulation	dairy streams	fluorescence	silica for MIR	off-spec liquid	surgical sensor

Source: outputs/EVAL.md, generated by run_eval.py --summary from the 14 committed reports.

SCALE · AND LIMITATIONS

Built to scale, honest about its limits

BUILT TO SCALE

Knowledge in seconds

A new document is searchable at once, no retraining.

No case-specific rules

General checks catch the traps for any customer.

Model-agnostic, API-first

Swap the model by config; the web app is one client.

Batch for queues

Whole lists answered unattended, with the same checks.

LIMITATIONS

A first assessment, not a validated set-up: the sources hold no measured results for a new customer.

Beyond AFS, abstracts only: gaps fall back to OpenAlex paper abstracts, never full papers, and count as weaker evidence.

LLM output varies between runs: the same case can get a different but defensible decision status.

Vision transcriptions can misread the image-only slides; they are committed so a reviewer can check them.

Run state lives in memory: a restart loses runs in progress (saved reports are safe), so we run one instance.

NEXT Persistent job queue and storage · several instances · expert feedback on each claim · enquiries from email or CRM

THANK YOU

Sourced. Checked. Ready for the expert.

Live demo · afsnationalaihackathon.vercel.app

Code · github.com/Adarshpandey2003/afs_national_ai_hackathon

